Tamanjyot Singh

(Official documents: Tamanjyot)

The University of Texas at Dallas, Richardson, TX, USA

tamanjyot@utdallas.edu singhtaman.github.io Tamanjyot singhtaman

Bio

Aerospace engineer pursuing a **doctorate in physics** with specialization in astrophysics. Interested in understanding **black hole spin** dynamics and their implications. Currently working as a research assistant at the University of Texas at Dallas under the supervision of **Dr. Michael Kesden**. My thesis work focuses on (i) **inclination distributions in tidal disruption events** by spinning supermassive black holes and their observational implications, (ii) **precessional and nutational** effects on the gravitational wave signals from inspiralling stellar mass binary black hole, and (iii) **degeneracies between lensed and precessing** gravitational wave signals for current LIGO sensitivities.

Education

Ph.D. **The University of Texas at Dallas** — Richardson, TX, USA
Physics — January 2021 - Ongoing
Thesis: *The Black Hole Ballet*: Spin and precession effects on gravitational waves and tidal disruption Events — *(expected May 2026)*
Advisor: Dr. Michael Kesden

M.S. **The University of Texas at Dallas** — Richardson, TX, USA
Physics — January 2021 - December 2023
GPA: 3.93/4.0

B.Tech. **Punjab Engineering College (Deemed to be University)** — Chandigarh, India
Aerospace Engineering with minor specialization in Mathematics — August 2016 - June 2020
GPA: 9.2/10.0, Rank 2
Major project: Computational study of flame propagation in a micro-scale channel and applications of micro-combustors in thermo-electric power generation.

Experience

Research assistant — Richardson, TX, USA
Department of Physics, The University of Texas at Dallas — May 2022 - Ongoing

- **Precession and Nutation in gravitational waves**: Investigating the effects of spin dynamics like precession and nutation in binary black hole systems on the gravitational wave emission.
- **Lensing vs Precession in gravitational waves**: Distinguishing between a lensed and precessing gravitational wave signal.
- **Tidal disruption events**: Exploring inclination biases induced by spin of the supermassive black hole in stellar disruptions by the supermassive black holes.

Teaching assistant — Richardson, TX, USA
Department of Physics, The University of Texas at Dallas — January 2021 - May 2022

- **Physics Laboratory I - Mechanics**: Introductory physics laboratory course.

Research Intern — Mumbai, India
Combustion Research Laboratory, Indian Institute of Technology, Bombay — January 2019 - April 2019

- **Combustion characteristics and effect of hydrogen content on syngas-air mixtures**: Formulated an empirical model to predict laminar burning velocity of multi-component gaseous mixtures at elevated pressures and temperatures.

IAF (Indian Air Force) Intern — Chandigarh, India
3rd Base Repair Depot (3BRD), Air Force Station — June 2018

Publications

Published †:

†"Distribution of orbital inclinations for tidal disruption events by Kerr black holes" — February 2024
Tamanjyot Singh, Michael Kesden
10.1103/PhysRevD.109.043016

Talks

Contributed	**Team for EXtragalactic AStrophysics (TEXAS) – Partnerships in Astronomy & Astrophysics Research and Education (PAARE) Annual Symposium 2025**, The University of Texas at Dallas, Richardson, TX	April 2025
Contributed	**APS Global Physics Summit 2025**, Anaheim, CA	March 2025
Contributed	**Fall meeting 2024, Texas Section APS**, Southern Methodist University, Dallas, TX	October 2024
Invited Visit	Center for Computational Astrophysics, Flatiron Institute, New York City, NY	May 2024
Contributed	**APS April meeting 2024**, Sacramento, CA	April 2024
Contributed	**Team for EXtragalactic AStrophysics (TEXAS) – Partnerships in Astronomy & Astrophysics Research and Education (PAARE) Annual Symposium 2024**, University of North Texas, Denton, TX	March 2024
Contributed	**10th Gulf Coast Gravity Meeting 2024**, University of Texas, Austin, TX	March 2024
Contributed	**International Conference on Gravitation and Cosmology 2023**, Indian Institute of Technology, Guwahati, Assam, India	December 2023
Contributed	**Fall meeting 2023, Texas Section APS**, Angelo State University, San Angelo, TX	October 2023
Contributed	**Spring meeting 2023, Texas Section APS**, Texas A&M University, Commerce, TX	March 2023

Awards

NSM Conference Travel Award — March 2025
School of Natural Sciences and Mathematics at The University of Texas at Dallas, TX

APS April meeting DGRAV Travel Award — April 2024
Division of Gravitational Physics, APS

Margie Renfrow travel award — March 2024
Department of Physics, The University of Texas at Dallas, TX

Teaching Assistant award 2022 — December 2022
Department of Physics, The University of Texas at Dallas, TX

PECOSA Sydney Prize(Merit Based - Awarded to 3 out of 700+ students) — September 2018
Punjab Engineering College, Chandigarh

1st position in the Mathematics Quiz — September 2016
Punjab State Council for Science and Technology, Chandigarh

99.3%ile in JEE (Mains) 2016 (among 1,500,000+ candidates all over India) — April 2016

International Rank 76 in International Olympiad of Science 2013 — November 2013
(Gold Medal + *Special Achievement Certificate*)

Workshops and Course Certificates

Participant in **Propects in Theoretical Physics 2025** — July 2025
Institute for Advanced Study, Princeton, NJ

Participant in **Entering Mentoring** — July 2024
The University of Texas at Dallas, Richardson, TX

Course Certificate for **Astro 101 : Black Holes** — June 2020
University of Alberta on Coursera.

Host for workshop on **Gravitational lensing** (Undergraduate level) — October 2019
Punjab Engineering College, Chandigarh

Participant in **ASTROWIN**, Winter School on Astronomy 2019 — February 2019
Birla Museum, Hyderabad

Course Certificate for **Particle Physics: an Introduction** — July 2018
University of Geneva on Coursera.

Course Certificate for **Understanding Einstein: The Special Theory of Relativity** — July 2017
Stanford University on Coursera.

Mentoring

I have had pleasure of working with many undergraduate and near-peer mentees over the years. Following is a list of names and roles of mentees:

Benjamin McCallip	Summer REU student 2025 *(Undergrad student)*
Tien Nguyen	Partnerships in Astronomy & Astrophysics Research and Education (PAARE) fellow 2023-24 *(Post-baccalaureate student)*
Maya Basu	Summer REU student 2024 *(Undergrad Student)*
Tinatin Tsiskaridze	Summer exchange student 2024 *(Grad Student)* (from Tblisi State University, Georgia)
Niel Shah	Summer REU student 2023 *(Undergrad Student)*
Deven Linthicum	Summer REU student 2023 *(Undergrad Student)*
Allison Powell	Summer REU student 2022 *(Undergrad Student)*

Leadership

Journal Club host: Hosting and leading weekly journal club meetings with astrophysics students and faculty across DFW in a joint effort between the University of Texas at Dallas, University of North Texas, Denton, and University of Texas at Arlington. — August 2024 - Ongoing

Graduate Students in Physics (GSP) Vice President: Representing graduate student interests in the department. Assisting President in maintenance and workings of the organization. — May 2023 - July 2024
Standout: Procured a department-wide Overleaf license.

Graduate Students in Physics (GSP) Astronomy committee head: Graduate student liaison for all things telescope, hosting star-watch and lunar eclipse parties. Responsible for maintenance of the existing equipment and inventory along with buying new equipment. — March 2022 - February 2024